Chow-variety dimension lower bounds on the size of variety-evasive subspace families

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Editorial note. Faithful editable transcription of the actual complete CC BY2024 journal main lower-bound proof and necessary original context. Original theorem/page references retained; classical geometry facts are standard prerequisites with the original author-provided statements/citations. No proof substituted or mathematical correction made. Native TeX retrieval has remote connection timeouts; actual journalPDF/page already retained, so this record is based directly on the licensed journal text. Independent lower-bound mechanism counted once, without splitting degree cases or classical Chow-dimension support. All production/extraction/validation solely on Yupeng.

Allowed standard prerequisites (editorial).

- Projective algebraic varieties, Grassmannians/Plucker coordinates, Chow embeddings and the standard dimension formula for Chow varieties. Original definitions and citedTheorems3.5/3.7/4.1/4.4 retained, with author-provided explanatory argument. Classical results and projective hyperplane dimension fact cited rather than recursively reproved.

Original ambient variety and evasion definitions (journal pp.3–6)

Let $\mathbb{F}$ be an algebraically closed field. An affine n-space $\mathbb{A}^n$, as a set, is simply defined to be the vector space $\mathbb{F}^n$. We also need the notion of a projective n-space, denoted by $\mathbb{P}^n$, which is (intuitively) the set of lines passing through the origin 0 of $\mathbb{A}^{n+1}$. Formally, it is defined to be the quotient set $(\mathbb{A}^{n+1} \setminus \{0\})/\sim$, where $\sim$ is the equivalence relation defined by scaling, i.e., $u \sim v$ if $u = cv$ for some nonzero scalar $c \in \mathbb{F}$. An (affine) subvariety $\mathcal{V} \subseteq \mathbb{A}^n$ is the set of common zeros of a set of n-variate polynomials over $\mathbb{F}$. Similarly, a (projective) subvariety $\mathcal{V} \subseteq \mathbb{P}^n$ is the set of common zeros of a set of homogeneous $(n+1)$-variate polynomials over $\mathbb{F}$, where we represent each element of $\mathbb{P}^n$ as an $(n+1)$-tuple in $\mathbb{A}^{n+1}$. In this paper, a variety refers to a subvariety of a projective or affine space, and is said to be irreducible if it cannot be written as a union of finitely many proper subvarieties. (Varieties in this paper are not necessarily irreducible. They are often called algebraic sets in literature.) The dimension of a variety $\mathcal{V}$, denoted by $\dim(\mathcal{V})$, is intuitively the "degree of freedom" of picking a point in the variety. See Section 2.3 for its formal definition. In particular,

for a linear subspace $\mathcal{V} \subseteq \mathbb{A}^n$, the dimension of $\mathcal{V}$ as a variety is simply its linear-algebraic dimension.

Definition 1.1 (Evading). *Let $\mathcal{V}_1$ and $\mathcal{V}_2$ be irreducible subvarieties of $\mathbb{P}^n$ or $\mathbb{A}^n$. We say $\mathcal{V}_1$ evades $\mathcal{V}_2$ if*

$$\dim(\mathcal{V}_1 \cap \mathcal{V}_2) \leq \dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) - n,$$

where the dimension of an empty set is assumed to be $-\infty$. In particular, if $\dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) < n$, then $\mathcal{V}_1$ evades $\mathcal{V}_2$ iff $\mathcal{V}_1 \cap \mathcal{V}_2 = \emptyset$. More generally, suppose $\mathcal{V}_1$ is irreducible but $\mathcal{V}_2$ is possibly reducible. We say $\mathcal{V}_1$ evades $\mathcal{V}_2$ if it evades every irreducible component of $\mathcal{V}_2$.

Definition 1.2 (Subspace family). *For $0 \leq k \leq n$, a finite collection (i.e., multiset) of k-subspaces of $\mathbb{P}^n$ is called a (projective) k-subspace family on $\mathbb{P}^n$. Similarly, a finite collection of affine k-subspaces of $\mathbb{A}^n$ is called an affine k-subspace family on $\mathbb{A}^n$.*

Definition 1.3 (Variety evasive subspace family). *Let $\mathcal{F}$ be a family of subvarieties of $\mathbb{P}^n$ (resp. $\mathbb{A}^n$). Let $\mathcal{H}$ be a k-subspace family on $\mathbb{P}^n$ (resp. affine k-subspace family on $\mathbb{A}^n$) where $0 \leq k \leq n$. Then:*

- *We say $\mathcal{H}$ is $\mathcal{F}$-evasive if for every $\mathcal{V} \in \mathcal{F}$, there exists $W \in \mathcal{H}$ that evades $\mathcal{V}$.*
- *We say $\mathcal{H}$ is $(\mathcal{F}, \epsilon)$-evasive if for every $\mathcal{V} \in \mathcal{F}$, a random element $W \in \mathcal{H}$ evades $\mathcal{V}$ with probability at least $1 - \epsilon$.*

Definition 1.4 (degree). *The degree of an irreducible variety $\mathcal{V}$ in $\mathbb{P}^n$ (resp. $\mathbb{A}^n$) is the number of intersections of $\mathcal{V}$ with a general projective (resp. affine) subspace of codimension $\dim(\mathcal{V})$. Following Heintz (1983); Heintz & Schnorr (1980), we define the degree of a (possibly reducible) variety to be the sum of the degrees of its irreducible components.*

Original Plücker and Chow-form background (journal pp.27–30)

Grassmannians. Let $n \in \mathbb{N}$ and $k \in \{0, 1, \ldots, n-1\}$. The Grassmannian $\mathrm{G}(k+1, n+1)$ is the set of all $(k+1)$-dimensional linear subspaces of $\mathbb{A}^{n+1}$. By taking the quotient modulo scalars, it may also be identified with the set of all k-subspaces of $\mathbb{P}^n$, which we denote by $\mathrm{G}(k, n)$.

The Plücker embedding and Plücker coordinates. Consider a linear subspace $W \in \mathrm{G}(k+1, n+1)$. The simplest way of representing W is using a $(k+1) \times (n+1)$ matrix A over $\mathbb{F}$ such that W equals the row space of A. We call such a matrix A a generating matrix of W. For convenience, we also say A is a generating matrix of $\mathbb{P}W \in \mathrm{G}(k, n)$.

Definition 3.4 (Plücker embedding). *Define* $\phi : \mathrm{G}(k+1, n+1) \to \mathbb{P}^{\binom{n+1}{k+1}-1}$ *by*

$$\phi(W) = \left(\det A_{[k+1],S}\right)_{S \in \binom{[n+1]}{k+1}}$$

where A *is a generating matrix of* W*.*

The Plücker embedding embeds the Grassmannian $\mathrm{G}(k+1, n+1)$ in $\mathbb{P}^{\binom{n+1}{k+1}-1}$ as an irreducible projective subvariety, as stated by the following theorem. See, e.g., Fulton (1997); Harris (1992) for proofs.

Theorem 3.5. *The Plücker embedding* ϕ *is a well-defined injective map whose image is an irreducible projective subvariety of* $\mathbb{P}^{\binom{n+1}{k+1}-1}$*.*

The homogeneous coordinates $\left(\det A_{[k+1],S}\right)_{S \in \binom{[n+1]}{k+1}}$ of $\phi(W)$ are called the (primal) Plücker coordinates of W. Denote by $R := \mathbb{F}[X_S : S \in \binom{[n+1]}{k+1}]$ the homogeneous coordinate ring of $\mathbb{P}^{\binom{n+1}{k+1}-1}$. The irreducible projective subvariety $\phi(\mathrm{G}(k+1, n+1))$ is defined by a homogeneous prime ideal of R, which we denoted by I. Then R/I is the homogeneous coordinate ring of $\phi(\mathrm{G}(k+1, n+1))$. The ideal I contains precisely the polynomial relations that the Plücker coordinates need to satisfy. It is also known that I is generated by certain quadratic forms, known as the Plücker relations. See Fulton (1997); Harris (1992) for details.

Dual Plücker coordinates. Alternatively, we could represent a linear subspace $W \in \mathrm{G}(k+1, n+1)$ by an $(n-k) \times (n+1)$ matrix B over $\mathbb{F}$ whose rows specify the linear equations defining W. We call such a matrix B a parity check matrix of W. For convenience, we also say B is a parity check matrix of $\mathbb{P}W \in \mathrm{G}(k, n)$. The entries of B are called the dual Stiefel coordinates of W. This gives another embedding $\phi^\vee : \mathrm{G}(k+1, n+1) \to \mathbb{P}^{\binom{n+1}{n-k}-1} = \mathbb{P}^{\binom{n+1}{k+1}-1}$, defined by

$$\phi^\vee(W) = \left(\det B_{[n-k],S}\right)_{S \in \binom{[n+1]}{n-k}}.$$

The homogeneous coordinates $\left(\det B_{[n-k],S}\right)_{S \in \binom{[n+1]}{n-k}}$ of $\phi^\vee(W)$ are called the dual Plücker coordinates of W. Recall that we denote by $\mathrm{G}(k, n)$ the set of all k-subspaces of $\mathbb{P}^n$. By identifying $\mathrm{G}(k+1, n+1)$ with $\mathrm{G}(k, n)$ via $W \mapsto \mathbb{P}W$, we regard ϕ and $\phi^\vee$ as maps from $\mathrm{G}(k, n)$ to $\mathbb{P}^{\binom{n+1}{k+1}-1}$.

Definition 3.6 (Gelfand et al. 1994). *For an irreducible subvariety* $\mathcal{V} \subseteq \mathbb{P}^n$ *of dimension* $n-k-1$*, define the associated hypersurfaces* $Z_\mathcal{V}$ *of* $\mathcal{V}$ *as the set of* k*-subspaces intersecting* $\mathcal{V}$*, i.e.,*

$$Z_\mathcal{V} := \{W \in \mathrm{G}(k, n) : \mathcal{V} \cap W \neq \emptyset\}.$$

The term "associated hypersurface" is justified by the following theorem.

Theorem 3.7. *Let* $\mathcal{V} \subseteq \mathbb{P}^n$ *be an irreducible projective subvariety of dimension* $n-k-1$ *and degree* $d \in \mathbb{N}^+$*. Then there exists a nonzero homogeneous polynomial* $P_\mathcal{V} \in R = \mathbb{F}[X_S : S \in \binom{[n+1]}{k+1}]$ *of degree* d *such that* $\phi(Z_\mathcal{V})$ *is defined by* $P_\mathcal{V}$ *as a subvariety of* $\phi(\mathrm{G}(k, n))$*. That is,*

$$\phi(Z_\mathcal{V}) = \phi(\mathrm{G}(k, n)) \cap V(P_\mathcal{V}).$$

Moreover, $R_{\mathcal{V}} := P_{\mathcal{V}} + I \in (R/I)_d$ is uniquely determined by $\mathcal{V}$ up to scalars.

Theorem 3.7 is explicitly stated as Dalbec & Sturmfels (1995, Theorem 1.1 and Corollary 2.1). A proof can be found in Gelfand et al. (1994, Section 3.2). We briefly explain how to find a polynomial $P_{\mathcal{V}}$ satisfying Theorem 3.7: Firstly, it can be shown using the trick of dimension counting via incidence varieties that $\phi(Z_{\mathcal{V}})$ is an irreducible projective subvariety of the Grassmannian $\phi(\mathrm{G}(k,n))$ of codimension one (Gelfand et al. 1994, Section 3.2, Proposition 2.2). Secondly, the homogeneous coordinate ring R/I of the Grassmannian is known to be a unique factorization domain (Fulton 1997, Chapter 9). These two facts imply that the homogeneous ideal of R/I defining $\phi(Z_{\mathcal{V}})$ is a principal ideal. Choose $R_{\mathcal{V}}$ to be a generator of this principal ideal, which is unique up to scalars. Then lift $R_{\mathcal{V}} \in R/I$ to $P_{\mathcal{V}} \in R$.

Definition 3.8 (Chow form). *Let $\mathcal{V} \subseteq \mathbb{P}^n$ be an irreducible subvariety of dimension $n-k-1$ and degree $d \in \mathbb{N}^+$. Define the Chow form of $\mathcal{V}$ in Plücker coordinates, or simply the Chow form of $\mathcal{V}$, to be $R_{\mathcal{V}} \in (R/I)_d$ as in Theorem 3.7. More generally, for an equidimensional subvariety $\mathcal{V} = \bigcup_{i=1}^s \mathcal{V}_i \subseteq \mathbb{P}^n$ of dimension $n-k-1$ and degree d, where $\mathcal{V}_1, \ldots, \mathcal{V}_s$ are the irreducible components of $\mathcal{V}$, the Chow form of $\mathcal{V}$ is $R_{\mathcal{V}} := \prod_{i=1}^s R_{\mathcal{V}_i} \in (R/I)_d$. It is uniquely determined by $\mathcal{V}$ up to scalars.*

As a k-subspace intersects $\mathcal{V} = \bigcup_{i=1}^s \mathcal{V}_i$ iff it intersects some $\mathcal{V}_i$, we see from Theorem 3.7 that the Chow form $R_{\mathcal{V}}$ of an equidimensional projective subvariety $\mathcal{V}$ of dimension $n-k-1$ vanishes precisely at the set of k-subspaces that intersect $\mathcal{V}$.

Theorem 1.10 (journal pp.8,43–44). *Let $n, d \in \mathbb{N}^+$ and $k \in \{0, 1, \ldots, n-1\}$. Let $\mathcal{F}$ be the family of equidimensional projective subvarieties of $\mathbb{P}^n$ of dimension $n-k-1$ and degree at most d. Suppose $\mathcal{H}$ is an $\mathcal{F}$-evasive k-subspace family on $\mathbb{P}^n$. Then*

$$|\mathcal{H}| \geq \begin{cases} (n-k)(k+1)+1 & \text{if } d = 1, \\ \max\left\{ d(n-k)(k+1)+1, \binom{d+n-k}{d} + (n-k+1)k \right\} & \text{if } d > 1. \end{cases}$$

In particular, $|\mathcal{H}|$ is superpolynomial in n when $n-k = n^{\Omega(1)}$ and $d = \omega(1)$.

Original Section 4: Lower Bound (journal pp.41–44)

We prove Theorem 1.10 in this section. The main tool is the notion of Chow varieties, which parameterize projective subvarieties. More precisely, they parametrize a generalization of projective subvarieties, called (effective) algebraic cycles on a projective space.

Algebraic cycles. An algebraic r-cycle (or simply r-cycle) on $\mathbb{P}^n$ is a formal linear combination $\mathcal{D} = \sum c_i \mathcal{V}_i$ of finitely many irreducible subvarieties $\mathcal{V}_i \subseteq \mathbb{P}^n$ of dimension r, where the coefficients c_i are integers. The degree of $\mathcal{D}$ is $\deg(\mathcal{D}) := \sum c_i \deg(\mathcal{V}_i)$. The support of $\mathcal{D}$ is $\mathrm{supp}(\mathcal{D}) := \bigcup_{c_i \neq 0} \mathcal{V}_i$. An r-cycle is effective if all its coefficients are nonnegative. Denote by $C(r,d,n)$ the set of all effective r-cycles of degree d on $\mathbb{P}^n$.

Chow varieties. Let $k \in \{0, 1, \ldots, n-1\}$ and $r = n - k - 1$. The definition of Chow forms naturally extends to effective r-cycles. Namely, for an effective r-cycle $\mathcal{D} = \sum_{i=1}^{r} c_i \mathcal{V}_i$ of degree d on $\mathbb{P}^n$, define the Chow form of $\mathcal{D}$ to be $R_{\mathcal{D}} := \prod_{i=1}^{r} R_{\mathcal{V}_i}^{c_i}$. Note that $R_{\mathcal{D}}$ is a vector in $(R/I)_d$ and is uniquely determined by $\mathcal{D}$ up to scalars. Write $[R_{\mathcal{D}}]$ for the point in $\mathbb{P}(R/I)_d$ represented by $R_{\mathcal{D}}$. Then we have map $\psi : C(r, d, n) \to \mathbb{P}(R/I)_d$, given by

$$\psi : \mathcal{D} \mapsto [R_{\mathcal{D}}],$$

called the Chow embedding of $C(r, d, n)$. Indeed, it embeds $C(r, d, n)$ in $\mathbb{P}(R/I)_d$ as a projective subvariety, as stated by the following theorem.

Theorem 4.1 (Chow & van der Waerden 1937). *The map ψ is injective and its image is Zariski-closed.*

A proof can also be found in Gelfand et al. (1994, Chapter 4). We identify $C(r, d, n)$ with its image under ψ and view it as a projective variety. This variety is called the Chow variety of effective r-cycles of degree d on $\mathbb{P}^n$.

Example 4.2. *Let V be the subspace of homogeneous polynomials in $\mathbb{F}[X_1, \ldots, X_{n+1}]$ of degree d. Then $C(n-1, d, n)$ is simply the projective space $\mathbb{P}V$ (see Example 3.9).*

Example 4.3. *$C(r, 1, n)$ is the Grassmannian $\mathrm{G}(r+1, n+1)$ (or $\mathrm{G}(r, n)$) embedded in $\mathbb{P}^{\binom{n+1}{r+1}-1} = \mathbb{P}^{\binom{n+1}{k+1}-1}$ via $\phi^\vee$ (see Example 3.10).*

The dimension of Chow varieties. When $d = 1$, the Chow variety $C(r, d, n)$ is just the Grassmannian $\mathrm{G}(r+1, n+1)$ (see Example 4.3) and its dimension is well known to be $(r+1)(n-r)$ (Harris 1992). When $d > 1$, the dimension of $C(r, d, n)$ was determined by Azcue (1992) and independently by Lehmann (2017). We state their result as follows.

Theorem 4.4 (Azcue 1992; Lehmann 2017). *For $d > 1$ and $0 \le r < n$, the dimension of $C(r, d, n)$ is*

$$\max\left\{ d(r+1)(n-r), \binom{d+r+1}{r+1} - 1 + (r+2)(n-r-1) \right\}.$$

This theorem was previously proved by Eisenbud & Harris (1992) for the special case $r = 1$.

Remark 4.5. *To prove Theorem 1.10, we only need a lower bound for the dimension of the Chow variety, which is much easier to prove than Theorem 4.4. Indeed, it is not difficult to see that $d(r+1)(n-r)$ is the dimension of the space of unions of d r-subspaces of $\mathbb{P}^n$, and $\binom{d+r+1}{r+1} - 1 + (r+2)(n-r-1)$ is the dimension of the space of degree-d hypersurfaces in $(r+1)$-subspaces of $\mathbb{P}^n$.*

Lower bound via dimension counting. We now restate Theorem 1.10 and prove it using a dimension counting argument.

Proof of Theorem 1.10. Consider an arbitrary k-subspace $W \in \mathcal{H}$. We may think of each point in $\mathbb{P}(R/I)_d$ as a homogeneous polynomial of degree d in Plücker coordinates modulo scalars and the ideal I of Plücker relations. We know Plücker coordinates always satisfy the Plücker relations. So it makes sense to talk about if a point in $\mathbb{P}(R/I)_d$ vanishes at $\phi(W)$ or not, as it does not depend on the choice of the homogeneous polynomial representing this point. Note that the constraint of $p \in \mathbb{P}(R/I)_d$ vanishing at $\phi(W)$ is a linear equation in the homogeneous coordinates of p. So the set of points in $\mathbb{P}(R/I)_d$ vanishing at $\phi(W)$ is a hyperplane of $\mathbb{P}(R/I)_d$, which we denote by H_W. Let $r = n-k-1$. Assume $|\mathcal{H}| \leq \dim(C(r,d,n))$. Then we have

$$C(r,d,n) \cap \bigcap_{W \in \mathcal{H}} H_W \neq \emptyset$$

since taking the intersection with a hyperplane reduces the dimension of a projective subvariety by at most one (Lemma 2.7 or Lemma 2.9). So there exists an effective r-cycle $\mathcal{D} = \sum_{i=1}^{s} c_i \mathcal{V}_i \in C(r,d,n)$, where $c_1, \ldots, c_s > 0$, such that $\psi(\mathcal{D}) = [R_{\mathcal{D}}]$ vanishes at $\phi(W)$ for all $W \in \mathcal{H}$. Let $\mathcal{V} = \operatorname{supp}(\mathcal{D}) = \bigcup_{i=1}^{s} \mathcal{V}_i$. Note $\mathcal{V} \in \mathcal{F}$ since $\deg(\mathcal{V}) \leq \deg(\mathcal{D}) = d$. For all $W \in \mathcal{H}$, we know $R_{\mathcal{D}} = \prod_{i=1}^{s} R_{\mathcal{V}_i}^{c_i}$ vanishes at $\phi(W)$, or equivalently, $R_{\mathcal{V}} = \prod_{i=1}^{s} R_{\mathcal{V}_i}$ vanishes at $\phi(W)$. This implies $\mathcal{V} \cap W \neq \emptyset$ for all $W \in \mathcal{H}$. As $\mathcal{V} \in \mathcal{F}$, this contradicts our assumption about $\mathcal{H}$. We conclude

$$|\mathcal{H}| \geq \dim(C(r,d,n)) + 1.$$

The dimension of $C(r,d,n)$ is $(r+1)(n-r)$ when $d = 1$ and is given by Theorem 4.4 when $d > 1$. Plugging in $r = n-k-1$ proves the theorem. □

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